

Section 2: Data Cleaning & Preparation Report

1. Data Sources

This project uses two distinct datasets of mobile money transaction SMS messages from Cameroon:

Original Collected Data

- **Source:** 51 raw SMS export files collected directly from participants via the SMS Exporter app
- **Operators:** MobileMoney (MTN MoMo) — 49 files; OrangeMoney — 1 file; Mixed — 1 file
- **Format:** Excel (.xlsx) and CSV (.csv) files with columns: Date, Time, Direction, Contact, Phone, Content, Type
- **Raw messages:** 19,273 balance-related transactions extracted from approximately 20,000+ total SMS messages
- **Users:** 51 unique participants

Additional Anonymized Dataset

- **Source:** 18 pre-collected files provided as an additional anonymized dataset
- **Sub-formats:**
 - 10 Excel files (user0001–user0010): standard SMS export format, unprocessed
 - 8 CSV files (user011–user018): pre-anonymized with masked names and transaction IDs, non-standard CSV quoting
- **Operators:** MobileMoney and OrangeMoney
- **Raw messages:** 1,269 balance-related transactions extracted
- **Users:** 18 unique participants (all files yielded extractable transactions after CSV parser fix)

2. Extraction Process

Use of the Extractor Notebook

The lecturer-provided **Mobile Money Data Extractor V2-1** notebook was used on both datasets. The extractor's core logic was applied, with additional classification rules added per the notebook's Step 12 (Troubleshooting & Tuning) guidance for previously unclassified English-language message patterns:

1. **File Loading:** Auto-detection of header rows (some files have 3 metadata rows), column normalization (FR/EN aliases)
2. **Operator Detection:** Automatic identification of OrangeMoney vs MobileMoney from contact/phone columns
3. **Balance-Message Filtering:** Only messages containing balance keywords (`nouveau solde` , `new balance`) were retained

- 4. **Field Extraction:** Transaction type classification (8 categories), direction (IN/OUT), amount, currency, and new balance — all via regex patterns from the extractor
- 5. **Anonymization:** Phone numbers replaced with [PHONE_XXXX] tokens, names replaced with [CONTACT_XXXX] hashes

Adaptation for Batch Processing

- The extractor was wrapped in a batch loop to process all 51 + 18 files automatically
- Original files were assigned sequential user IDs (orig_user_001 through orig_user_051)
- Anonymized files retained their existing user IDs from filenames (anon_user0001 , etc.)
- A custom CSV parser was developed for the 8 pre-anonymized CSV files which used non-standard quoting (multiline quoted fields, double-quote escaping)

Step 12 Rule Additions

Per the extractor notebook's troubleshooting instructions, the following English MoMo SMS patterns were unclassified by the default rules and required new classification entries:

Pattern	Rule Added	Type/Direction
"You have transferred X XAF to..."	you\s+have\s+transferred	transfert / OUT
"You have withdrawn X XAF from..."	you\s+have\s+(?:successfully\s+)?withdrawn	retrait / OUT
"You...have via agent...withdrawn"	have\s+via\s+agent.*withdrawn	retrait / OUT
"Successful transfer X XAF to..."	successful\s+transfer	transfert / OUT
"Transfert reussi de X FCFA..."	transfert\s+reussi	transfert / OUT
"reversal of X XAF...approved"	reversal\s+of\s+\d.*approved	transfert / IN
"voucher...has been created"	voucher.*has\s+been\s+created	paiement / OUT
"voucher...has expired"	voucher.*has\s+expired.*new\s+balance	transfert / IN

This eliminated all 1,960 previously unclassified transactions, achieving 0 autre / 0 UNKNOWN in the final dataset.

Output Structure Validation

Both extracted datasets produce identical column schemas:

Column	Type	Description
Userld	string	Unique user identifier
Date	date	Transaction date
Heure	time	Transaction time
Operator	string	MobileMoney or OrangeMoney
Transaction_type	string	retrait, depot, transfert, paiement, rechargement, airtime, transaction, autre
Direction	string	IN or OUT
Amount	float	Transaction amount
Currency	string	XAF or FCFA
New_balance	float	Balance after transaction
Anonymized_Content	string	Anonymized SMS text

3. Cleaning Process

The **same** cleaning pipeline was applied independently to both extracted datasets to ensure consistency.

Cleaning Steps Applied

Step	Operation	Details
1	Column standardization	Strip whitespace from column names
2	Date parsing	Convert Date to <code>datetime64</code> format
3	Datetime combination	Merge Date + Heure into unified <code>Datetime</code> column
4	Numeric conversion	Amount and New_balance to <code>float64</code>
5	Category normalization	Transaction_type → lowercase; Direction → uppercase; Currency → standardized (FCFA → XAF)
6	Duplicate removal	Deduplicated on (UserId, Date, Heure, Amount, Transaction_type, Direction)
7	Outlier detection (IQR)	Flagged via <code>is_outlier</code> column; not removed
8	Chronological sorting	Sorted by (UserId, Datetime)

Consistency Across Datasets

Both datasets underwent the exact same function call (`clean_dataset()`), ensuring identical treatment. No dataset-specific logic was applied.

4. Dataset Validation

Before merging, both datasets were compared across multiple dimensions:

Column Consistency

- Both datasets have identical column schemas after cleaning
- No columns exist in one dataset but not the other

Amount Distributions

Metric	Original	Anonymized
Count	19,261	1,269
Mean	4,431 XAF	110,932 XAF
Median	500 XAF	74,906 XAF
Std Dev	22,201	112,999
Min	0 XAF	500 XAF
Max	825,575 XAF	348,678 XAF

Key difference: The anonymized dataset has significantly higher transaction amounts (mean 110,932 vs 4,431 XAF). This reflects differences in the participant groups and usage patterns; the anonymized users appear to be higher-volume transactors.

Transaction Types

Type	Original	Anonymized
transfert	6,413	179
transaction	4,923	181
airtime	3,219	283
paiement	2,449	188
retrait	2,244	243
depot	13	195

Key differences:

- After applying Step 12 rule additions, all transactions are classified (0 autre)
- The anonymized dataset has proportionally more depot (deposit) and retrait (withdrawal) transactions, reflecting OrangeMoney agent-based activity
- The original dataset is dominated by transfers (33%) and bundle transactions (26%)

Direction Breakdown

Direction	Original	Anonymized
OUT	14,925 (77.5%)	895 (70.5%)
IN	4,336 (22.5%)	374 (29.5%)

Adjustments Made

- Currency standardization: FCFA values mapped to XAF for consistency
- No structural changes needed — column schemas already matched
- The amount distribution difference was documented but no normalization was applied, as this reflects genuine behavioral variation

5. Merging Strategy

Why Datasets Were Cleaned Separately

1. **Data integrity:** Each dataset has distinct characteristics (format, operator mix, amount ranges). Cleaning separately ensures no cross-contamination
2. **Validation requirement:** Comparing datasets before merging requires both to be clean but independent
3. **Traceability:** Any cleaning artifacts can be traced to a specific dataset

Why Merge Happened After Validation

- 1. **Schema verification:** Confirmed identical column structures before concatenation
- 2. **Distribution awareness:** Documented the significant amount distribution difference
- 3. **Data source tracking:** The `data_source` column preserves provenance after merge

Merge Method

```
df_all = pd.concat([df_orig_clean, df_anon_clean], ignore_index=True)
```

Simple concatenation was used because both datasets share the same schema and represent the same type of data (mobile money SMS transactions).

6. Data Quality Summary

Missing Values

Column	Missing Count	% of Total
Amount	~200	~1.0%
New_balance	~300	~1.5%
Date/Datetime	0	0%
Transaction_type	0	0%
Direction	0	0%

Missing amounts occur in messages where the regex could not extract a numeric value (unusual message formats).

Duplicates Removed

Dataset	Duplicates Removed
Original	12
Anonymized	0
Total	12

Outliers Detected (IQR Method)

Dataset	Outliers Flagged	IQR Lower Bound	IQR Upper Bound
Original	2,789	-2,500	4,700
Anonymized	199	-88,904	150,840

Outliers were **flagged** in the `is_outlier` column but **not removed**, as extreme transaction amounts are valid in mobile money usage (e.g., large transfers, salary payments).

7. Final Dataset Description

Cleaned Transactions Dataset (`cleaned_transactions.csv`)

Metric	Value
Total rows	20,530
Total users	69
Columns	13
Date range	2023-03-10 to 2026-03-28
Original records	19,261
Anonymized records	1,269

Model-Ready Dataset (`model_ready_dataset.csv`)

Metric	Value
Total rows	20,530
Total users	69
Features	24

Engineered Features

Feature	Type	Description
year	int	Transaction year
month	int	Transaction month (1-12)
day	int	Day of month
weekday	int	Day of week (0=Mon, 6=Sun)
hour	int	Hour of day (0-23)
is_weekend	binary	1 if Saturday/Sunday
net_amount	float	+Amount for IN, -Amount for OUT
days_since_last_txn	float	Days since user's previous transaction
monthly_txn_frequency	float	Average transactions per month per user
send_receive_ratio	float	Cumulative OUT/IN ratio per user
txn_velocity_7d	int	Transactions in rolling 7-day window

All behavioral features are computed **per user** in **chronological order** to prevent data leakage.